



Taar.

A phone can tell you whether a wire is carrying current — from three centimetres away, touching nothing, with the radio off.

Goutham Pakala · Geetha Tigulla
Phase 1 idea submission

2–3×

NO CURRENT

9–16×

CURRENT FLOWING

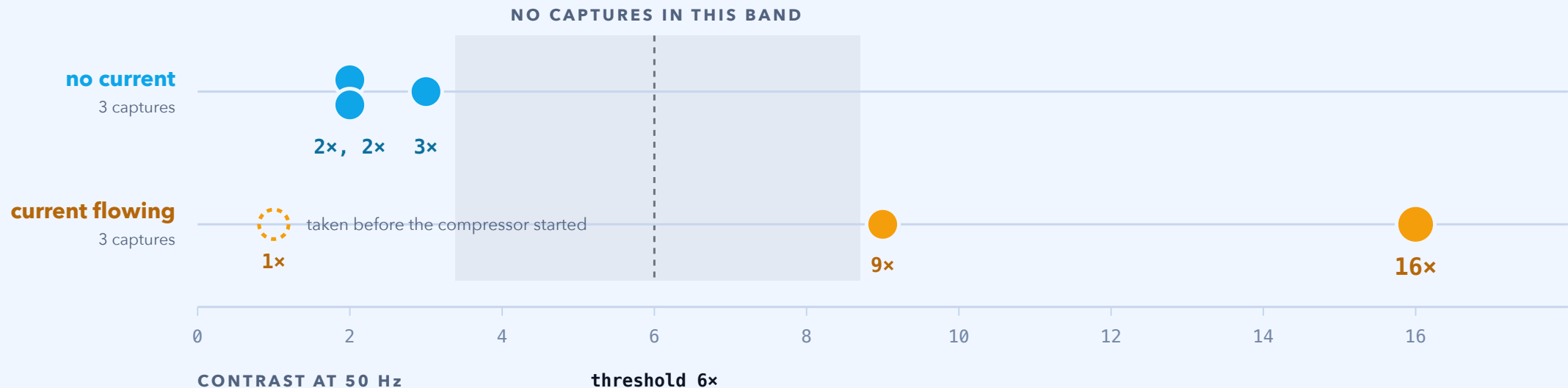
44/44

CHECKS PASS

Two populations, and a gap between them.

3×

SEPARATION, NO OVERLAP



The threshold moved to 6x because of this plot. The previous value, chosen from simulation, sat close enough to the idle readings that a dead circuit could read as live.

One capture in the lower lane read 1x, taken before the compressor had started. **We set it aside on an explanation formed after the fact, and the separation depends on that being right.**

Same phone. Same cable. One switch.

Two pairs, minutes apart. The phone was taped down and never touched; the only thing that changed in the room was a switch at the wall.

APPLIANCE OFF · NOTHING MOVED

Mains detected	3x contrast · 0.27
Field amplitude	0.05 μT

Mains detected	2x contrast · 0.15
Field amplitude	0.05 μT

COMPRESSOR RUNNING · SAME POSITION

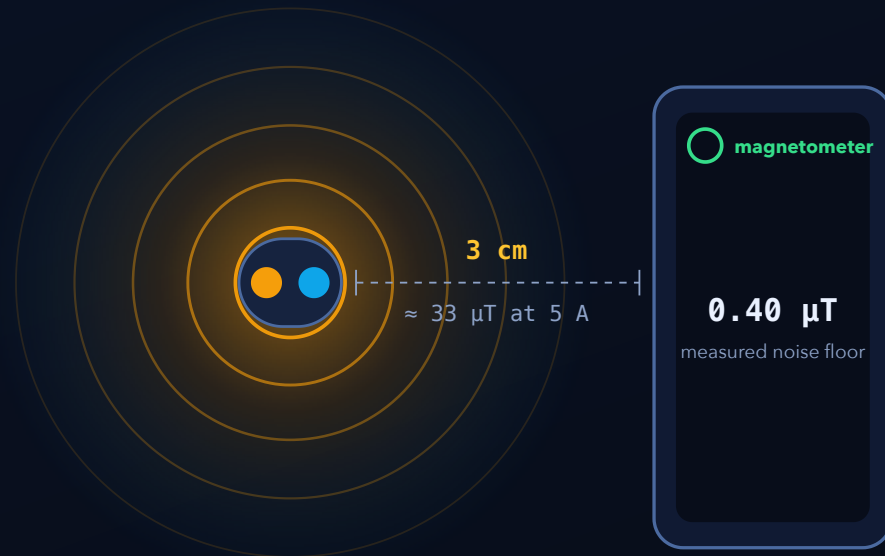
Mains detected	16x contrast · 0.64
Field amplitude	0.14 μT

Mains detected	9x contrast · 0.50
Field amplitude	0.10 μT

Both numbers come off the device, not from a model. Field amplitude 0.05 μT idle against 0.10–0.14 loaded, on a noise floor measured at 0.40.

What this does **not** show is a sensitivity figure. One appliance, roughly 1–1.5 A, through a cable whose live and neutral largely cancel.

WHY IT WORKS



● live ● neutral their fields largely cancel, which is why the signal is small

TWO PHYSICAL FACTS NOBODY HAS POINTED A PHONE AT

$$B = \mu_0 I / 2\pi r$$

A conductor broadcasts its current

Five amperes at three centimetres produce about 33 μT . Earth's field is 45. **The signal is the same order as the planet** – it was never too small to see.

$$100 \text{ Hz} = 2 \times \text{mains}$$

An arc broadcasts its fault

A series arc re-ignites twice per cycle, modulating its noise at 100 Hz. Against a ballast hum, speech, a sleeve and a motor, a plain energy detector fired on **every one**. Ours held its 5% design point.

WHO IT IS FOR

The instruments cost more than the firm earns in a month.

A clamp meter, a thermal camera and an arc-fault analyser are what make a loose connection visible before it starts a fire. For a two-person electrical firm that is a capital outlay they cannot justify – so the tools stay with large contractors, and the buildings that most need checking never get checked.

Every one of those electricians already carries a magnetometer and a microphone.

WHAT IT REPLACES, PARTLY

Is this circuit dead before I touch it? yes

Is it over its rating? yes, once calibrated

Is anything arcing behind the plate? attempted

Cost to the electrician a phone they own

Taar reports a relative index, not amperes, unless the circuit has been calibrated against a known load.

Everything we got wrong, we got wrong in simulation.

The app passed every check we had written and still returned nothing on a real phone. These four only appeared once it was installed.

01	Two Android clocks	<code>SensorEvent.timestamp</code> counts deep sleep; <code>System.nanoTime()</code> does not. Comparing them ended every capture on its first sample.
02	Magnitude destroyed the signal	Reducing three axes to $\sqrt{x^2+y^2+z^2}$ hid the AC component behind Earth's static field. A 1 μ T signal moved it by 0.01.
03	Normalised by the wrong quantity	A hand-held capture drifts several μ T. That drift buried the 50 Hz term and reported a real signal as zero confidence.
04	A threshold we guessed	The arc floor was five times too small, and called a domestic fridge an arcing fault. Re-set from nine measured captures.

A simulation that had predicted any of these would have been a better simulation. It did not – which is the case for building early and putting it on the hardware.

Not a concept.

Pre-check

Measured, not requested. A reading is only as good as what the sensors actually delivered.

OK

Magnetometer present

OK

Sample rate
105.3 Hz measured · requested 99 Hz · 21 distinct phases

OK

Timing jitter
0.0% of the interval — uniform, but 21 phases is plenty

OK

Noise floor
0.40 μ T RMS

OK

Microphone
fell back to MIC — arc band may be attenuated

!!

Reference
none — record a baseline before measuring

PRE-CHECK · MEASURED, NOT REQUESTED

Logic checks passing	44 / 44
Debug APK, builds clean	9.3 MB
Network permissions in the manifest	0
Magnetometer rate achieved	105.3 Hz · 21 phases
Noise floor, two sessions	0.39 – 0.40 μ T
Faults in the catalogue, English & Telugu	6

Signal path

Per-axis sine fit at 50 Hz, detrended, amplitude as the vector sum. Lomb-Scargle contrast against 30-48 and 52-70 Hz. Scoring by median and MAD against the circuit's own reference.

On top of it

A rules table that shows its evidence. A nearest-centroid classifier **trained on the device** from the technician's labels, which rejects rather than guessing. No language model is built; one is planned for the build window.

SCOPE

What Taar does not do.

It is a triage aid, not a certification

It does not replace a licensed electrician, a calibrated clamp meter, an insulation tester or a statutory inspection. The app says so on the result screen, not in a settings page.

The acoustic half is designed, not validated

No real arc has been recorded. The magnetometer half rests on textbook physics and now on measurement; the arc half rests on neither yet, and we will say so on stage before a judge asks.

It reports an index, not amperes

Unless the circuit has been calibrated against a known load. A fabricated ampere figure is worse than no figure.

It refuses rather than guesses

With no reference, or an ill-conditioned fit, it returns a refusal. An electrician acting on a confident wrong reading is the worst outcome this project has, and the code is arranged so that outcome is unreachable.

Our pre-event work is a research spike on a separate branch, rebuilt from an empty scaffold inside the event window and never merged. The protocol is written down, and we will confirm it with an organiser at check-in.

Nothing in the room moves but a switch.

3x

APPLIANCE OFF · NOTHING MOVED

Mains detected	3x contrast · 0.27
Field amplitude	0.05 uT

ONE SWITCH

phone taped down,
never touched

16x

COMPRESSOR RUNNING · SAME POSITION

Mains detected	16x contrast · 0.64
Field amplitude	0.14 uT

Yesterday it was a claim. Today there are **six captures** behind it.

Taar · Against Default
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